

2)

$$f(x_1, x_2) = (x_1 - 3)^2 + x_1 x_2 + (x_2 - 3)^2$$

$$(1) \quad \frac{\partial f}{\partial x_1} = 2(x_1 - 3) + x_2 \quad \frac{\partial f}{\partial x_2} = x_1 + 2(x_2 - 3)$$

$$(2) \quad \nabla f = 0$$

$$2x_1 + x_2 = 6 \quad (1)$$

$$x_1 + 2x_2 = 6 \quad (2)$$

$$(3) \quad \text{from (1) } x_2 = 6 - 2x_1 \text{ and replace on (2)}$$

$$x_1 + 2(6 - 2x_1) = 6$$

$$x_1 + (12 - 4x_1) = 6$$

$$-3x_1 = -6$$

$$\Rightarrow x_1 = 2$$

$$\Rightarrow x_2 = 6 - 2(2) = 2$$

Stationary point: $x^{(2)} = (2, 2)$

(4) Hessian

$$H = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

(5)

Main first minor: $2 > 0$

Determinant: $2 \cdot 2 - 1 = 3 > 0$

therefore $x^{(2)} = (2, 2)$ is a local minimum